STALKS, SUPPORT AND RANGE OF TOR SHEAVES ON A NONSINGULAR VARIETY

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file intersection.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Editorial note. Occupation scan: combined claim_key + statement_anchor + generic-line scan over all registration files (active and withdrawn) found 0 hits for this chapter (is-occupied.json, 2026-10-14); consistent with the duty manager's six-check dispatch.

Source reference. Original lemma 0AZS: sources/stacks-chow/tags.txt, source lines 13835–13835. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 00P0: sources/stacks-chow/tags.txt, source lines 914–914. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 00OE: sources/stacks-chow/tags.txt, source lines 892–892. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 066P: sources/stacks-chow/tags.txt, source lines 7943–7943. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let X be a nonsingular variety. Let $\mathcal{F}$, $\mathcal{G}$ be coherent $\mathcal{O}_X$-modules. The $\mathcal{O}_X$-module $Tor_p^{\mathcal{O}_X}(\mathcal{F}, \mathcal{G})$ is coherent, has stalk at x equal to $Tor_p^{\mathcal{O}_{X,x}}(\mathcal{F}_x, \mathcal{G}_x)$, is supported on $Supp(\mathcal{F}) \cap Supp(\mathcal{G})$, and is nonzero only for $p \in \{0, \ldots, \dim(X)\}$.*

Proof. The result on stalks was discussed above and it implies the support condition. The Tor's are coherent by Lemma 0AZS. The vanishing of negative Tor's is immediate from the construction. The vanishing of Tor_p for $p > \dim(X)$ can be seen as follows: the local rings $\mathcal{O}_{X,x}$ are regular (as X is nonsingular) of dimension $\leq \dim(X)$ (Algebra, Lemma 00P0), hence $\mathcal{O}_{X,x}$ has finite global dimension $\leq \dim(X)$ (Algebra, Lemma 00OE) which implies that Tor-groups of modules vanish beyond the dimension (More on Algebra, Lemma 066P). □

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